

Birzeit University
Department of Electrical & Computer Engineering
ENCS5343 Computer Vision, First Semester, 2025/2026
Image-to-Image Retrieval System using Classical CV Techniques

1. Overview

Image retrieval is a fundamental problem in computer vision where the goal is to retrieve images from a large database that are visually similar to a given query image. Modern retrieval systems combine powerful visual feature extraction with efficient indexing methods to achieve accurate and fast search. In this project, we design a compact image-to-image retrieval pipeline that demonstrates key concepts in feature extraction, feature matching, similarity measurement, and evaluation. The system takes a query image, extracts discriminative features, compares them with stored image features in the database, and returns the top-k most similar images. The project emphasizes understanding both classical and modern feature representations.

2. Objectives

1. **Understand image retrieval fundamentals** including feature extraction, representation, and similarity scoring.
2. **Implement a complete retrieval pipeline** using classical methods.
3. **Choose an appropriate retrieval methodology.**
4. **Evaluate the retrieval system** using standard metrics such as mAP and Precision@K.

3. System Flowchart (Basic Pipeline)

Input Image → Preprocessing → Feature Extraction → Feature Database → Similarity Computation → Ranking → Retrieved Images

The image retrieval process begins with the user providing a query image. This image is first preprocessed, which may include resizing, normalization, or conversion to grayscale depending on the type of features to be extracted. In the feature extraction step, classical methods such as color features, local features such as SIFT or SURF. The extracted features are compared against a pre-computed feature database containing feature vectors for all images in the dataset. Similarity between the query and database images is computed using measures such as Euclidean distance or cosine similarity. Based on these similarity scores, the images are ranked, and the top-K most similar images are returned and displayed as the retrieval results.

4. Dataset

For this project, an appropriate dataset should contain multiple objects or scenes with repeated instances to test the ability to retrieve similar images. Recommended options:

- **Oxford5k Buildings** (classical instance retrieval, widely used)
- **Paris6k**
- **Caltech-101**: Object-category dataset with multiple instances per class.

5. Evaluation Measures

To assess retrieval quality, you will use:

1. **Precision@K (P@K)**: Measures the proportion of relevant images among the top-K retrieved results.
2. **Recall@K (R@K)**: Measures how many of all relevant images in the dataset appear within the top-K retrieved results.
3. **Mean Average Precision (mAP)**: The main metric for image retrieval. It computes:
 - Precision at each relevant retrieval
 - Average Precision (AP) for each query
 - $mAP = \text{mean}(AP \text{ over all queries})$
4. **Retrieval Time**: Measures system efficiency, especially important when working with large image databases.

6. Deliverables

- **Implementation Code**: A complete and well-organized implementation of the proposed image retrieval system.
- **Comprehensive Report**: The report should contain the following sections:
 - **Introduction**
 - **Detailed description of the proposed method**
 - **Experimental results**
 - **A comparison table highlighting performance relative to other related work**

7. Evaluation

- **Proposed Method – 60%**: Assessed based on correct implementation, **proper choice of methodology, contribution**, system performance, and how well the retrieval pipeline is designed and executed.
- **Report – 40%**: Evaluated on clarity, completeness, and organization. Must include an introduction, detailed method description, experimental results, comparison table with related work, and overall quality of presentation.